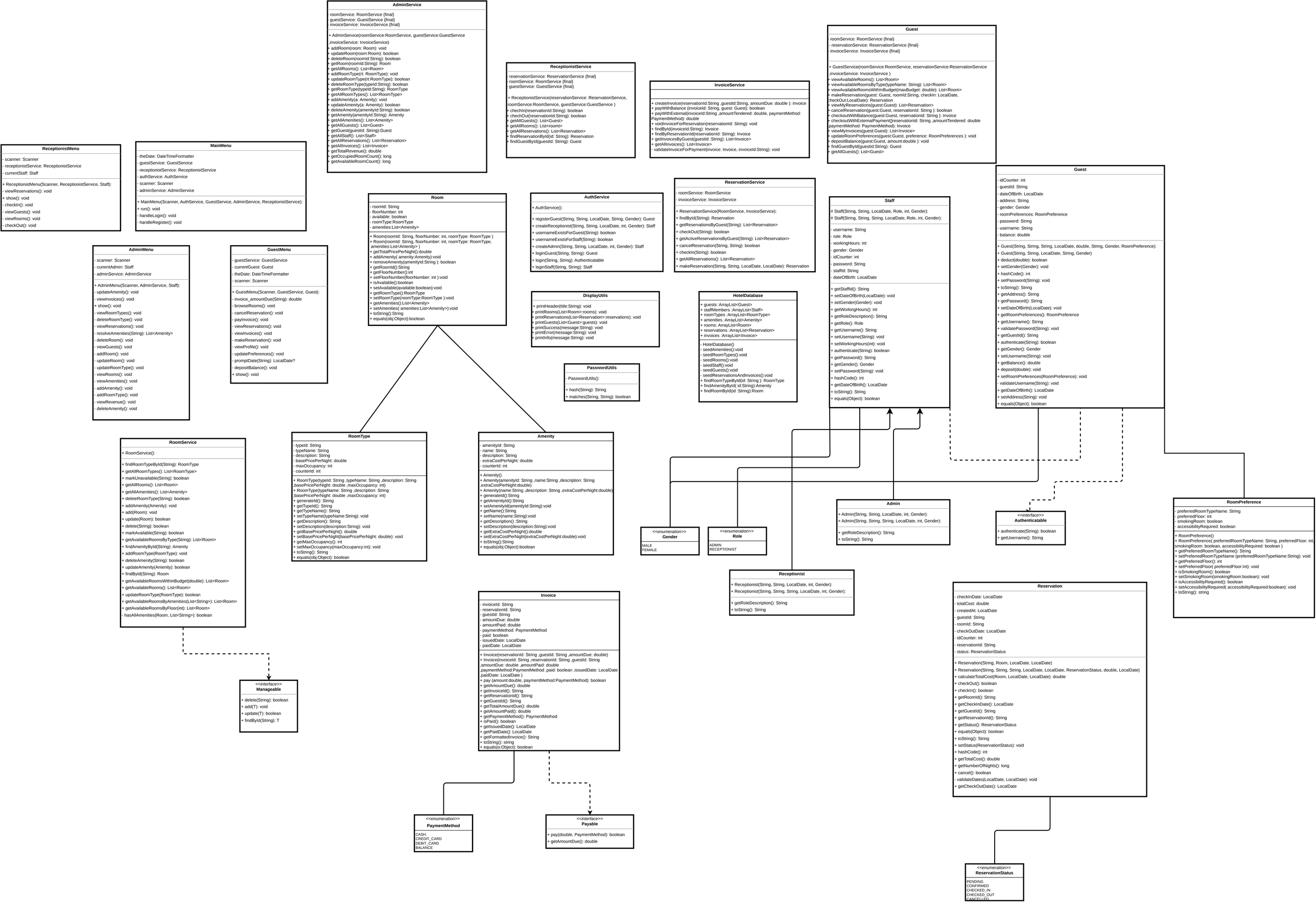
AdminService
- roomService: RoomService {final}
- guestService: GuestService {final}
- invoiceService: InvoiceService {final}
+ AdminService(roomService:RoomService, guestService:GuestService ,invoiceService: InvoiceService)
+ addRoom(room: Room): void
+ updateRoom(room:Room): boolean
+ deleteRoom(roomId:String): boolean
+ getRoom(roomId:String): Room
+ getAllRooms(): List<Room>
+ addRoomType(rt: RoomType): void
+ updateRoomType(rt:RoomType): boolean
+ deleteRoomType(typeId:String): boolean
+ getRoomType(typeId:String): RoomType
+ getAllRoomTypes(): List<RoomType>
+ addAmenity(a: Amenity): void
+ updateAmenity(a: Amenity): boolean
+ deleteAmenity(amenityId:String): boolean
+ getAmenity(amenityId:String): Amenity
+ getAllAmenities(): List<Amenity>
+ getAllGuests(): List<Guest>
+ getGuest(guestId: String):Guest
+ getAllStaff(): List<Staff>
+ getAllReservations(): List<Reservation>
+ getAllInvoices(): List<Invoice>
+ getTotalRevenue(): double
+ getOccupiedRoomCount(): long
+ getAvailableRoomCount(): long
ReceptionistService
- reservationService: ReservationService {final}
- roomService: RoomService {final}
- guestService: GuestService {final}
+ ReceptionistService(reservationService: ReservationService, roomService:RoomService, guestService:GuestService)
+ checkIn(reservationId:String): boolean
+ checkOut(reservationId:String): boolean
+ getAllGuests(): List<Guest>
+ getAllRooms(): List<roomt>
+ getAllReservations(): List<Reservation>
+ findReservationById(id: String): Reservation
+ findGuestById(guestId: String): Guest
InvoiceService
+ createInvoice(reservationId:String ,guestId:String, amountDue: double): Invoice
+ payWithBalance (invoiceId: String, guest: Guest): boolean
+ payWithExternal(invoiceId:String ,amountTendered: double, paymentMethod: PaymentMethod): double
+ voidInvoiceForReservation(reservationId: String): void
+ findById(invoiceId:String): Invoice
+ findByReservationId(reservationId: String): Invoice
+ getInvoicesByGuest(guestId :String): List<Invoice>
+ getAllInvoices(): List<Invoice>
- validateInvoiceForPayment(invoice: Invoice, invoiceId:String): void
Guest
- roomService: RoomService {final}
- reservationService: ReservationService {final}
- invoiceService: InvoiceService {final}
+ GuestService(roomService:RoomService, reservationService:ReservationService ,invoiceService: InvoiceService)
+ viewAvailableRooms(): List<Room>
+ viewAvailableRoomsByType(typeName: String): List<Room>
+ viewAvailableRoomsWithInBudget(maxBudget: double): List<Room>
+ makeReservation(guest: Guest, roomId:String, checkIn: LocalDate, checkOut:LocalDate): Reservation
+ viewMyReservations(guest:Guest): List<Reservation>
+ cancelReservation(guest:Guest, reservationId: String): boolean
+ checkoutWithBalance(guest:Guest, reservationId: String): Invoice
+ checkoutWithExternalPayment(reservationId: String ,amountTendered: double paymentMethod: PaymentMethod): Invoice
+ viewMyInvoices(guest:Guest): List<Invoice>
+ updateRoomPreferences(guest:Guest, preference: RoomPreferences): void
+ depositBalance(guest:Guest, amount:double): void
+ findGuestById(guestId:String): Guest
+ getAllGuests(): List<Guest>
ReceptionistMenu
- scanner: Scanner
- receptionistService: ReceptionistService
- currentStaff: Staff
+ ReceptionistMenu(Scanner, ReceptionistService, Staff):
- viewReservations(): void
+ show(): void
- checkIn(): void
- viewGuests(): void
- viewRooms(): void
- checkOut(): void
MainMenu
- theDate: DateTimeFormatter
- guestService: GuestService
- receptionistService: ReceptionistService
- authService: AuthService
- scanner: Scanner
- adminService: AdminService
+ MainMenu(Scanner, AuthService, GuestService, AdminService, ReceptionistService):
+ run(): void
- handleLogin(): void
- handleRegister(): void
Room
- roomId: String
- floorNumber: int
- available: boolean
- roomType:RoomType
- amenities:List<Amenity>
+ Room(roomId: String, floorNumber: int, roomType: RoomType)
+ Room(roomId: String, floorNumber: int, roomType: RoomType, amenities:List<Amenity>)
+ getTotalPricePerNight():double
+ addAmenity(amenity:Amenity):void
+ removeAmenity(amenityId:String): boolean
+ getRoomId():String
+ getFloorNumber():int
+ setFloorNumber(floorNumber: int):void
+ isAvailable():boolean
+ setAvailable(available:boolean):void
+ getRoomType():RoomType
+ setRoomType(roomType:RoomType):void
+ getAmenities():List<Amenity>
+ setAmenities(amenities:List<Amenity>):void
+ toString():String
+ equals(obj:Object):boolean
AuthService
+ AuthService():
+ registerGuest(String, String, LocalDate, String, Gender): Guest
+ createReceptionist(String, String, LocalDate, int, Gender): Staff
+ usernameExistsForGuest(String): boolean
+ usernameExistsForStaff(String): boolean
+ createAdmin(String, String, LocalDate, int, Gender): Staff
+ loginGuest(String, String): Guest
+ login(String, String): Authenticatable
+ loginStaff(String, String): Staff
ReservationService
- roomService: RoomService
- invoiceService: InvoiceService
+ ReservationService(RoomService, InvoiceService):
+ findById(String): Reservation
+ getReservationsByGuest(String): List<Reservation>
+ checkOut(String): boolean
+ getActiveReservationsByGuest(String): List<Reservation>
+ cancelReservation(String, String): boolean
+ checkIn(String): boolean
+ getAllReservations(): List<Reservation>
+ makeReservation(String, String, LocalDate, LocalDate): Reservation
Staff
Staff(String, String, LocalDate, Role, int, Gender):
Staff(String, String, String, LocalDate, Role, int, Gender):
- username: String
- role: Role
- workingHours: int
- gender: Gender
- idCounter: int
- password: String
- staffId: String
- dateOfBirth: LocalDate
+ getStaffId(): String
+ setDateOfBirth(LocalDate): void
+ setGender(Gender): void
+ getWorkingHours(): int
+ getRoleDescription(): String
+ getRole(): Role
+ getUsername(): String
+ setUsername(String): void
+ setWorkingHours(int): void
+ authenticate(String): boolean
+ getPassword(): String
+ getGender(): Gender
+ setPassword(String): void
+ hashCode(): int
+ getDateOfBirth(): LocalDate
+ toString(): String
+ equals(Object): boolean
Guest
- idCounter: int
- guestId: String
- dateOfBirth: LocalDate
- address: String
- gender: Gender
- roomPreferences: RoomPreference
- password: String
- username: String
- balance: double
+ Guest(String, String, String, LocalDate, double, String, Gender, RoomPreference):
+ Guest(String, String, LocalDate, String, Gender)
+ deduct(double): boolean
+ setGender(Gender): void
+ hashCode(): int
+ setPassword(String): void
+ toString(): String
+ getAddress(): String
+ getPassword(): String
+ setDateOfBirth(LocalDate): void
+ getRoomPreferences(): RoomPreference
+ getUsername(): String
+ validatePassword(String): void
+ getGuestId(): String
+ authenticate(String): boolean
+ getGender(): Gender
+ setUsername(String): void
+ getBalance(): double
+ deposit(double): void
+ setRoomPreferences(RoomPreference): void
- validateUsername(String): void
+ getDateOfBirth(): LocalDate
+ setAddress(String): void
+ equals(Object): boolean
DisplayUtils
+ printHeader(title:String): void
+ printRooms(List<Room>:rooms): void
+ printReservations(List<Reservation>:reservations): void
+ printGuests(List<Guest>:guests): void
+ printSuccess(message:String): void
+ printError(message:String): void
+ printInfo(message:String): void
HotelDatabase
+ guests: ArrayList<Guest>
+ staffMembers :ArrayList<Staff>
+ roomTypes :ArrayList<RoomType>
+ amenities :ArrayList<Amenity>
+ rooms: ArrayList<Room>
+ reservations :ArrayList<Reservation>
+ invoices :ArrayList<Invoice>
- HotelDatabase()
- seedAmenities():void
- seedRoomTypes():void
- seedRooms():void
- seedStaff():void
- seedGuests():void
- seedReservationsAndInvoices():void
+ findRoomTypeById(id: String): RoomType
+ findAmenityById(id:String):Amenity
+ findRoomById(id :String):Room
PasswordUtils
- PasswordUtils():
+ hash(String): String
+ matches(String, String): boolean
AdminMenu
- scanner: Scanner
- currentAdmin: Staff
- adminService: AdminService
+ AdminMenu(Scanner, AdminService, Staff):
- updateAmenity(): void
- viewInvoices(): void
+ show(): void
- viewRoomTypes(): void
- deleteRoomType(): void
- viewReservations(): void
- resolveAmenities(String): List<Amenity>
- deleteRoom(): void
- viewGuests(): void
- addRoom(): void
- updateRoom(): void
- updateRoomType(): void
- viewRooms(): void
- viewAmenities(): void
- addAmenity(): void
- addRoomType(): void
- viewRevenue(): void
- deleteAmenity(): void
GuestMenu
- guestService: GuestService
- currentGuest: Guest
- theDate: DateTimeFormatter
- scanner: Scanner
+ GuestMenu(Scanner, GuestService, Guest):
- invoice_amountDue(String): double
- browseRooms(): void
- cancelReservation(): void
- payInvoice(): void
- viewReservations(): void
- viewInvoices(): void
- makeReservation(): void
- viewProfile(): void
- updatePreferences(): void
- promptDate(String): LocalDate?
- depositBalance(): void
+ show(): void
RoomService
+ RoomService():
+ findRoomTypeById(String): RoomType
+ getAllRoomTypes(): List<RoomType>
+ markUnavailable(String): boolean
+ getAllRooms(): List<Room>
+ getAllAmenities(): List<Amenity>
+ deleteRoomType(String): boolean
+ addAmenity(Amenity): void
+ add(Room): void
+ update(Room): boolean
+ delete(String): boolean
+ markAvailable(String): boolean
+ getAvailableRoomsByType(String): List<Room>
+ findAmenityById(String): Amenity
+ addRoomType(RoomType): void
+ deleteAmenity(String): boolean
+ updateAmenity(Amenity): boolean
+ findById(String): Room
+ getAvailableRoomsWithinBudget(double): List<Room>
+ getAvailableRooms(): List<Room>
+ updateRoomType(RoomType): boolean
+ getAvailableRoomsByAmenities(List<String>): List<Room>
+ getAvailableRoomsByFloor(int): List<Room>
- hasAllAmenities(Room, List<String>): boolean
RoomType
- typeId: String
- typeName: String
- description: String
- basePricePerNight: double
- maxOccupancy: int
- counterId: int
+ RoomType(typeId: String ,typeName: String ,description: String ,basePricePerNight: double ,maxOccupancy: int)
+ RoomType(typeName: String ,description: String ,basePricePerNight: double ,maxOccupancy: int)
+ generateId(): String
+ getTypeId(): String
+ getTypeName(): String
+ setTypeName(typeName:String): void
+ getDescription(): String
+ setDescription(description:String): void
+ getBasePricePerNight(): double
+ setBasePricePerNight(basePricePerNight: double): void
+ getMaxOccupancy(): int
+ setMaxOccupancy(maxOccupancy:int): void
+ toString(): String
+ equals(obj:Object): boolean
Amenity
- amenityId: String
- name: String
- description: String
- extraCostPerNight: double
- counterId: int
+ Amenity()
+ Amenity(amenityId: String ,name:String ,description: String ,extraCostPerNight:double)
+ Amenity(name:String ,description: String ,extraCostPerNight:double)
+ generateId():String
+ getAmenityId():String
+ setAmenityId(amenityId:String):void
+ getName():String
+ setName(name:String):void
+ getDescription() :String
+ setDescription(description:String):void
+ getExtraCostPerNight():double
+ setExtraCostPerNight(extraCostPerNight:double):void
+ toString():String
+ equals(obj:Object):boolean
<<interface>>
Manageable
+ delete(String): boolean
+ add(T): void
+ update(T): boolean
+ findById(String): T
Invoice
- invoiceId: String
- reservationId: String
- guestId: String
- amountDue: double
- amountPaid: double
- paymentMethod: PaymentMethod
- paid: boolean
- issuedDate: LocalDate
- paidDate: LocalDate
+ Invoice(reservationId: String ,guestId: String ,amountDue: double)
+ Invoice(invoiceId: String ,reservationId: String ,guestId: String ,amountDue: double ,amountPaid: double ,paymentMethod:PaymentMethod ,paid: boolean ,issuedDate: LocalDate ,paidDate: LocalDate)
+ pay (amount:double, paymentMethod:PaymentMethod): boolean
+ getAmountDue(): double
+ getInvoiceId(): String
+ getReservationId(): String
+ getGuestId(): String
+ getTotalAmountDue(): double
+ getAmountPaid(): double
+ getPaymentMethod(): PaymentMethod
+ isPaid(): boolean
+ getIssuedDate(): LocalDate
+ getPaidDate(): LocalDate
+ getFormattedInvoice(): String
+ toString(): string
+ equals(o:Object): boolean
<<enumeration>>
PaymentMethod
CASH,
CREDIT_CARD
DEBIT_CARD
BALANCE
<<interface>>
Payable
+ pay(double, PaymentMethod): boolean
+ getAmountDue(): double
<<enumeration>>
Gender
MALE
FEMALE
<<enumeration>>
Role
ADMIN
RECEPTIONIST
Admin
+ Admin(String, String, LocalDate, int, Gender):
+ Admin(String, String, String, LocalDate, int, Gender):
+ getRoleDescription(): String
+ toString(): String
<<interface>>
Authenticatable
+ authenticate(String): boolean
+ getUsername(): String
Receptionist
+ Receptionist(String, String, LocalDate, int, Gender):
+ Receptionist(String, String, String, LocalDate, int, Gender):
+ getRoleDescription(): String
+ toString(): String
RoomPreference
- preferredRoomTypeName: String
- preferredFloor: int
- smokingRoom: boolean
- accessibilityRequired: boolean
+ RoomPreference()
+ RoomPreference(preferredRoomTypeName: String, preferredFloor: int, smokingRoom: boolean, accessibilityRequired: boolean)
+ getPreferredRoomTypeName(): String
+ setPreferredRoomTypeName (preferredRoomTypeName:String): void
+ getPreferredFloor(): int
+ setPreferredFloor(preferredFloor:int): void
+ isSmokingRoom(): boolean
+ setSmokingRoom(smokingRoom:boolean): void
+ isAccessibilityRequired(): boolean
+ setAccessibilityRequired(accessibilityRequired:boolean): void
+ toString(): string
Reservation
- checkInDate: LocalDate
- totalCost: double
- createdAt: LocalDate
- guestId: String
- roomId: String
- checkOutDate: LocalDate
- idCounter: int
- reservationId: String
- status: ReservationStatus
+ Reservation(String, Room, LocalDate, LocalDate)
+ Reservation(String, String, String, LocalDate, LocalDate, ReservationStatus, double, LocalDate)
+ calculateTotalCost(Room, LocalDate, LocalDate): double
+ checkOut(): boolean
+ checkIn(): boolean
+ getRoomId(): String
+ getCheckInDate(): LocalDate
+ getGuestId(): String
+ getReservationId(): String
+ getStatus(): ReservationStatus
+ equals(Object): boolean
+ toString(): String
+ setStatus(ReservationStatus): void
+ hashCode(): int
+ getTotalCost(): double
+ getNumberOfNights(): long
+ cancel(): boolean
- validateDates(LocalDate, LocalDate): void
+ getCheckOutDate(): LocalDate
<<enumeration>>
ReservationStatus
PENDING
CONFIRMED
CHECKED_IN
CHECKED_OUT
CANCELLED